

PROJECT REPORT

AgroSense AI

Agricultural Recommendation System — Project Report

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Repository: <https://github.com/MohanPrabu018-K/Agro-Sense-AI-> (MIT License)
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1. Executive Summary

AgroSense AI is a Flask-based smart-agriculture web application that supports practical farm decisions through three core capabilities: crop recommendation, fertilizer planning, and image-based disease screening. It combines a trained Random Forest crop model, rule-based nutrient planning, heuristic leaf-image analysis, user authentication, prediction history, PDF report generation, and an admin dashboard in a single local web system.

2. Problem Statement

Farmers, agriculture students, and early-stage agri-tech users struggle to select the right crop for a field, judge whether nutrient levels are balanced, and tell whether a plant image indicates disease risk — without a single decision-support interface. Manual reasoning depends heavily on experience. AgroSense AI turns soil, climate, and image inputs into usable agricultural guidance.

3. Objectives

- Recommend suitable crops from soil and climate inputs using a trained ML model.
- Plan fertilizer requirements using crop-specific NPK targets.
- Screen leaf images for disease risk and reject non-plant images.
- Store prediction history with downloadable PDF reports and an admin usage view.

4. System Architecture

The system follows a Flask monolith architecture: Jinja-rendered templates for the frontend, Flask routes in app.py for the backend, a local JSON persistence file for runtime data, a pickled Random Forest crop model for inference, and Pillow-based heuristic image screening.

Layer	Implementation
Frontend	HTML + CSS + JavaScript, Jinja templates (12 pages)
Backend	Python + Flask (auth, prediction, dashboard, admin, PDF routes)
ML / Data	scikit-learn Random Forest, pandas; model/crop_model.pkl + metadata
Image analysis	Pillow heuristic color-pattern screening
Storage	Local JSON store (users, predictions, feedback)
External	Open-Meteo API for weather auto-fill

5. Key Features

- Crop recommendation from N, P, K, temperature, humidity, pH, and rainfall.
- Soil health scoring (Excellent / Good / Moderate / Needs Improvement), crop care guide, and profitability estimate.
- Fertilizer planner comparing current NPK against crop-specific targets with quantity guidance.
- Disease screening across five risk categories with confidence, urgency, signals, and preventive guidance.
- Invalid-image rejection, prediction history, PDF field reports, feedback collection, admin dashboard, English/Tamil label toggle.

6. Data & Model

Training data: dataset/crop_recommendation.csv — 2,200 rows across 22 crop classes (N, P, K, temperature, humidity, pH, rainfall, label), generated reproducibly by tools/generate_crop_dataset.py. train_model.py validates schema, encodes labels, trains a RandomForestClassifier, saves the (model, encoder) artifact, and prints validation accuracy and a classification report. Model metadata is tracked in model/model_metadata.json.

7. Data Flow

- Offline: dataset CSV → train_model.py → model/crop_model.pkl.
- Runtime: user input → Flask validation → crop inference / fertilizer rules / image screening → stored prediction → result page / dashboard / PDF.

8. Testing & Verification

- Verified: home and health routes (HTTP 200), registration, login, admin access, anonymous-prediction redirect to login, crop/fertilizer/disease predictions, non-plant image rejection, and PDF route.

9. Results

The application delivers one workflow-oriented platform covering crop suitability, fertilizer guidance, and disease support, with evaluator-friendly traceability: dataset, training script, model artifact, and metadata are all included in the repository.

10. Challenges & Future Scope

- Challenge: establishing model ownership trust and handling invalid uploads — addressed with a full dataset-to-artifact pipeline and non-plant rejection.
- Out of scope by design: certified agronomic diagnosis, cloud deployment, deep-learning disease classification, and market-price integration.
- Future: CNN/ViT disease classifier, larger real-world datasets, email verification, analytics charts, Dockerized deployment, PostgreSQL migration.

11. Repository & Reproduction

Repository: <https://github.com/MohanPrabu018-K/Agro-Sense-AI-> — run run_app.ps1 (or run_app.bat) from the project root and open <http://127.0.0.1:5000>. Demo/admin access details are documented in the repository.